Formula Name **KETAMINE HCl / MORPHINE SO4 / BUPIVACAINE HCL TOPICAL SPRAY 4/4/0.75% SOLN**

Initials	AP	**Quantity**	50 MLS	**Schedule**	2	**Days to Exp**	35	**Date of Formula**	8/9/2023
Cost	$12.85	**Cost Per Unit**	**$0.26**	**AWP**	$0.00	**AWP Per Unit**	**$0.00**	**Last Review**	1/6/2016

Chemical Ingredients

Chemical Name	Form	Qty	Units	QS	Est Cost	Ha	Note
KETAMINE HCL USP	PWD	2.0000	GMS	☐	$4.35		SOLUBILITY: WATER (FS) ALCOHOL (S) MAXIMUM SOLUBILITY IS 200MG/ML WHITE CRYSTALLINE POWDER WITH A CHARACTERISTIC COLOR SOLUTIONS MAY BE CLEAR TO SLIGHTLY GREEN USED IN ANESTHESIA MAY CAUSE AMNESIA
MORPHINE SULFATE (PENTAHYDRATE) USP	PWD	2.0000	GMS	☐	$6.00		SOLUBILITY: 1:16 WATER(S), 1:570 ALCOHOL(SL)
BUPIVACAINE HYDROCHLORIDE MONOHYDRATE USP	PWD	0.3750	GMS	☐	$2.48		SOLUBILITY: 1:25 WATER(S), 1:8 ALCOHOL(FS)
SODIUM CHLORIDE GRANULAR USP/NF	PWD	0.2800	GMS	☐	$0.02		1:2.8 WATER(FS), ALCOHOL (SL)
PRESERVED WATER (PARABENS)	LIQU	0.0000	MLS	☑	$0.00		SEE FORMULA FOR PRESERVED WATER (PARABEN)

Therapy

Classification(s) Preserved Aqueous

Description Clear colorless solution

Disease State(s)

Instructions

A. Calculations/Special Considerations:
Bupivacaine Hydrochloride USP Monohydrate may have a water determination between 4 – 6% and an assay between 98.5 – 101.5% calculated on the anhydrous basis. It is important that you obtain a Certificate of Analysis for your specific lot number to verify the percentage of water determination and assay (anhydrous basis).
**The amount of Bupivacaine Hydrochloride in the final calculation is generally 3-7% above the emperic calculation.
Empiric calculation for this compound yields 0.75g/100mL or 7.5mg/mL

Calculations refer to CoA for Bupivacaine Hydrochloride USP Lot 195042: Water determination% = 5.3%, Assay=100.6%
1. Subtract the water determination % from 100%. 100% - __5.3__ % = amount of pure Bupivacaine HCl ___94.7___ %
3. Divide 7.5mg/mL by the answer from step 2. 7.5mg/ml / ___94.7___% = Required conc. of Bupivacaine HCl___7.920__ mg/mL
4. Divide the required conc. of Bupivacaine HCl__7.920_ mg/mL (answer from step 3) by the assay purity percentage listed on the CoA. ___7.920__mg/mL / __100.6__ % = The required concentration of Bupivacine HCl needed for this compound Lot __7.873__mg/mL
5. Multiply required concentration of Bupivacaine needed for this compound Lot (answer from step 4) by the total volume. Required conc. of Bupivacaine HCl needed for this compound Lot __7.873__ mg/mL x total volume of compound _______mL = Amount of Bupivacaine HCl powder needed to weigh = _________ g

B. Equipment Required:
Weigh boats; beakers; graduated cylinders, scales; stir plate; stir bar; funnel; tape, calculator, metered spray pump

C. Method of Preparation:
1. Calculate required amounts and fill out compounding and control usage logs
2. Measure out ingredients. Determine the Bupivacaine HCl amount by the completing the calculation above
3. Add 80% of final volume of preserved water to beaker and add stir bar
4. Dissolve Ketamine HCl, Morphine SO4, Bupivacaine HCl, and Sodium Chloride in Preserved Water while stirring.
5. QS to final volume with Preserved water in class A graduated cylinder
6. Have compound certified pharmacist perform final verification
7. Pour into metered pump amber spray bottle - Capacity is 60mL

8. Label with RX label, lot number and expiration barcode,
9. Tape Labels

D. Labeling:
Label with lot number and expiration barcode and ingredient(s) label.

Add prepack label if compound lot made to stock for multiple Rxs
Add Rx auxillary label if compound lot made to order for a unique Rx

________(place label here)________

E. Description of Finished Product:
Clear colorless solution

F. Quality Control Procedures:
Physical appearance of compound verified by compounding certified pharmacist
Volume of final product is within 5% of intended batch size

Pharmacist signature:________________________

G. Packaging Container:
Package in metered pump amber spray bottle (60mL capacity)

H. Storage Requirements
Store at controlled room temperature (68°–77 °F)
Maximum BUD - 35 days

I. Formula Source & BUD Testing:
Formulation - PCCA 5026 & 7527
BUD - USP 795

INGREDIENT WEIGHT #1 INGREDIENT WEIGHT #2 INGREDIENT WEIGHT #3

INGREDIENT WEIGHT #4